We will look at the games and applications based on Java later on in this chapter.

5.2.2 BREW

We have dealt with Java and its role in mobile telephony, but there is one more platform for application programming—BREW, which stands for Binary Runtime Environment for Wireless. It is air-interface independent, i.e. it supports GSM/GPRS, UTMS and CDMA. However, when BREW was first introduced, it was developed solely for CDMA handsets.

BREW provides solutions for wireless application development in a variety of languages such as C/C++, Java, XML etc. Device configuration and billing/payment are also facilitated by BREW. The complete solution package includes the BREW SDK for application developers, client software and porting tools for device manufacturers, and the BREW Delivery System (managed by operators).

There is a misconception that BREW is in competition with Java. The truth is that BREW supports all programming languages including Java. The BREW client acts an extended platform for other programming languages and environments (such as JVM). Any type of browser (HTML, WAP, cHTML) can run on BREW as an application. This flexibility helps incorporate a wide range of applications.

BREW runs between the application and the wireless device's chip operating system. This enables a programmer to develop applications without having to code for system interface or understand any wireless application. That's not all. The most important advantage of this platform is that it eliminates the hassles of modifying an application with every new phone model or network, thus decreasing the application development cost. All these are true only for a BREW enabled phone (according to Qualcomm, the creator of BREW).